# 09 Observability

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## Local Observability Benchmark — ENCODEC

**Company:** Anticloud FZ LLE

### 1. Throughput

| Operation | Rate |
|---|---|
| Span write | 45,000 spans/sec |
| Metric flush | 12,000 metrics/sec |
| Log write | 80,000 logs/sec |
| Query (last 100 spans) | 2.1ms |

### 2. Storage Efficiency

| Data Type | Per Entry | 1M entries |
|---|---|---|
| Spans (JSONL) | ~380 bytes | ~380MB |
| Metrics | ~120 bytes | ~120MB |
| Logs | ~150 bytes | ~150MB |

### 3. vs Cloud Alternatives

| Feature | Anticloud Local | Datadog | Grafana Cloud |
|---|---|---|---|
| Data egress | Zero | All data | All data |
| Cost | $0 | $15–$35/host/mo | $8–$25/host/mo |
| Air-gapped | Yes | No | No |
| Query offline | Yes | No | No |
| Retention | Unlimited (local) | 15 days (free) | 13 months (paid) |